CS6120: Natural Language Processing (NLP)

A field of artificial intelligence that enables computers to understand, interpret, and respond to human language in a meaningful way.

Note: This syllabus is subject to change. Students will be promptly notified of any changes.

Time and Location: Wednesdays, 1:00 PM – 4:20 PM, Room 136/137 Portland

There will be weekly synchronous meetings.

Instructor: Prashant Mittal

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Course Overview

This course provides a broad view of modern Natural Language Processing (NLP), discusses methodological foundations, and issues of practical implementation, use, and techniques for assessing performance. This is an applied **case-study-based course** that requires a successful demonstration of competency in foundational concepts in machine learning using **Python programming**.

NLP focuses on the interaction between computers and humans through natural language. NLP enables computers to understand, interpret, and generate human languages meaningfully and helpfully. The applications are wide-ranging and include tasks such as sentiment analysis, which can gauge public opinion on products or political issues; machine translation, which enables real-time translation between different languages; text summarization, providing concise summaries of lengthy documents; and speech recognition, used in virtual assistants like Siri and Alexa. Additionally, NLP powers chatbots in customer service, aids in information retrieval from large datasets, and plays a vital role in many other areas where understanding and generation of human language are required.

The course is tailored for graduate students offering study of text analytics and language processing. It encompasses foundational concepts like tokenization and Bag-of-Words structured form, word embedding representations, and modern language models and techniques such as transformers. If time permits, students will explore corporate contextual implementations ethical considerations in NLP, and engage in hands on assignments. The curriculum emphasizes analytical skills and model development. It does not focus on deployment of models, as that topic is related to data engineering. Assessments include team based biweekly tasks and a capstone project, culminating in a well-rounded education. The course ensures that participants are well-equipped to analyze, interpret, and generate human language using cutting-edge models and techniques.

Open source tools such as Kaggle, ChatGPT (+ Bard, Claude), github copilot are embedded/ encouraged to use.

Course Learning Outcomes

At the completion of this course, students should be able to:

- Understand core concepts, methodologies, and challenges in text analytics and language processing
- Acquire hands-on experience using various tools and techniques to analyze, interpret, and generate human language.
- Gain deep insights into Large Language Models (LLMs), embeddings, and other state-of-the-art models, understanding their strengths, limitations, and appropriate application contexts.
- Approach complex textual datasets with a strategic analytical mindset, using both classical and machine learning-based methods to extract meaningful insights.
- Develop a foundational understanding that prepares them for continuous learning, staying updated with the latest advancements in the field.

Required Book

Hands on Machine Learning with scikit-learn (3rd edition)
This is the main text for the course. The text can be accessed via NU library here:
You may need to register with O'reilly media if you haven't already registered.

Course Schedule

One biweekly group assignment where students are expected to submit **fully executable programs, results, and a short-written summary including strengths / weaknesses of the analysis**.

The following agenda is subject to change due to everchanging fluid nature of NLP concepts. Students will be notified in advance.

Module	Description	Assignment	Due Date
A	**Introduction to Natural Language Processing** - What is NLP? - Overview of Language Models - Applications and Challenges in NLP - Setting up Environment (Python, NLTK, Spacy, Gensim, Keras, PyTorch) - Examples of how I have used NLP	A0: Regular Expressions • Skills survey • Read Chapters 1 to 4 • Come prepared to discuss ML fundamentals	TBD
A	**Text Processing, Language Modeling** - Tokenization, Stemming & Lemmatization, Text Normalization - N-grams, Language Model Estimation, Smoothing Techniques, Zips' law - Rule-Based and Statistical POS Tagging - Bag-of-Words (BoW), Vectorization, TFIDF - Rule-Based and Machine Learning Approaches - Unsupervised Learning techniques	A1 – Clustering of Physician Notes	TBD

A	**Machine Learning** - Foundations of Machine Learning - Text Classification Techniques: Statistical models, SVM, Tree based models		
B	**Deep Learning** - The ANN framework - ndarrays - RNN and CNN frameworks	A2: Classification of documents	
C	**Sentiment Analysis and Topic Modeling** - Lexicon Based approach - ML approach - Latent Dirichlet Allocation (LDA) - Correlated Topic Modeling (CTM)		
C	**Word, sentence, document representation (Embeddings)** - Word Embeddings (Word2Vec, GloVe) - Abstractive Summarization with Seq2Seq - Sentence and document embeddings - Importing and Utilizing Embeddings - Text Summarization	A3	
D	**Transformers** - Tour of Huggingface Ecosystem - A tour of Transformers - Challenges with Transformers - Transfer Learning	A4	
D	**Transformer Anatomy** - Encoder Decoder framework - Attention based architecture - Transformer Attention Mechanism		
D	**Fine Tuning Transformers** - XLM-RoBERTa - Performance Measures - Error Analysis	A5	
E	**Text Generation and Summarization**	Final Project Discussion 1	
F	**Large Language Models and Recent Trends** - LLMs like GPT-3, GPT-4 - Understanding Bias in Language Models - Transfer Learning in NLP - Explainable AI	Final Project Discussion 2	
G	Final Presentation		

Grades

Item	Percentage	Total
HW assignments A0	5%	5%
HW Assignment A1, A2	10% each	20%
HW assignments A3, A4	15% each	30%
HW assignment A5	10%	10%
Final Presentation	25%	25%
Attendance and Participation	10%	10%

Letter grades will be assigned on the following scale: 93%+ A, 90-92% A-, 87-89 B+, 83-87% B, 80-82% B-, 77-79 C+, 73-77% C, 70-72% C-,60-70% D, 0-59% F.

Office Hours

To ensure students receive additional academic support outside regular class hours and foster a collaborative learning environment, the TA will set aside time every week specifically for office hours. During this time, students can discuss class-related topics, seek clarifications, or delve deeper into specific subject areas.

Students must come prepared with specific questions or topics they wish to discuss. They are encouraged to email a summary of their questions or topics to the TA in advance. This ensures that the limited time is efficiently utilized and allows the tutor to assist effectively. Students are expected to respect the allocated time slots and only extend their consultation within their designated time, especially if other students are waiting. Punctuality is essential.

Policy on ChatGPT (and other chatbot tools)

Students are authorized to use ChatGPT (and other chatbot tools) as a supplementary learning and coding tool. The intent is to foster deeper understanding, encourage research, and support coding activities. Most class **lectures will allocate a specific duration for students to utilize such tools for research purposes**. During this time, students pose queries to chatbot tools to seek clarifications, examples, or additional information on the topic under discussion. Students will reconvene to discuss their findings post-research, allowing for collaborative learning. This is intended to spark engaging discussions, enabling students to learn from chatbot tools and their peers. **Once the designated research time concludes, students are expected to cease all interactions with the chatbot tool. Lingering on the platform or using it beyond the allotted time is prohibited**.

Upon completing the research segment, students must actively engage in ongoing class activities, discussions, and exercises without distractions. Failure to adhere to this policy, especially lingering on the computer beyond the permitted research time, **may result in penalties**.

Students must use ChatGPT responsibly, ensuring that the platform is utilized solely for academic and research purposes related to the course. Any attempt to misuse or leverage the platform for non-academic gains will be considered a violation of this policy.

Assignment Details

Each homework will have detailed instructions specific to the assignment. The student should carefully read the description of the homework and what is asked from each problem in the assignment. For programming assignments, you are allowed to work together, but are expected to write your own code.

Final Presentation:

In lieu of a traditional final, you are required to make a formal presentation at the end of the semester. The details of the assignment and related data will be discussed toward the end of the semester. Your group will submit a project proposal, a written report, and also deliver a presentation during the final class. Attendance at that session is mandatory.

Classroom Conduct:

To create and preserve a classroom atmosphere that optimizes teaching and learning, all participants share a responsibility in creating a civil and non-disruptive forum for the discussion of ideas. Students are expected to conduct themselves at all times in a manner that does not disrupt teaching or learning.

Your comments to others must be constructive and free from harassing statements. You are encouraged to disagree with other students and the instructor, but such disagreements need to be respectful and be based upon facts and documentation, rather than prejudices and personalities. The instructor reserves the right to interrupt conversations that deviate from these expectations.

Repeated unprofessional or disrespectful conduct may result in a lower grade or more severe consequences.

Attendance and Tardiness:

Attendance is mandatory. Missing a class not only causes you to miss the information disseminated in that lecture but can cause you to miss important information in regards to quizzes and assignments. If you miss class, it is your responsibility to catch up on the material you missed including obtaining notes from a classmate and watching the recorded lectures. If you have circumstances that can prevent you from being in class on time, please let me know. Classes will be held on campus at The Roux each week, but the lectures will also be streamed via Zoom, and recorded. Any deviation in the delivery of the lecture will be announced through Canvas.

Late Policy:

You are allowed 5 "**slip-days**" throughout the semester. This means that you may turn in an assignment late, where each day it is late will reduce your number of slip-days by 1. Once you are out of slip-days, if you turn in the assignment late, you will earn a 0 for that assignment. You may not use slip-days on quizzes, or for presentations. You may not re-allocate slip-days; once they are used, they are gone.

Communication:

The easiest way to get in touch with me is to send me an email. I will check my Northeastern email account regularly and respond promptly. Please note that occasionally I may not respond until the next business day.

Academic Honesty:

As with all other courses at Northeastern, you are expected to adhere to the university's academic integrity policy (http://www.northeastern.edu/osccr/academic-integrity).

- You may not copy someone else's code or answer under any circumstances. This includes from online sources.
- You may discuss and work through problems with others, and, in fact, I encourage you to do so. However, ultimately, your solution that you turn in must be your original work.

  If you have had a substantive discussion of any homework or programming solution with a classmate, then be sure to cite them in your report. If you are unsure of what constitutes "substantive", then ask us or err on the side of caution. You will not be penalized for working together. You must not copy answers or code from another student either by hand or electronically.

  Disability: If you have any kind of disability that can affect your performance in this class, please let me know privately through email or finding me on campus for a private chat.

  Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that substantiates a disability and demonstrates a current significant limitation. Ac- commodations are provided based on an evaluation of the information provided by students, and their clinicians, on a case-by-case basis. Students should provide documentation to the DRC at their earliest convenience to allow for sufficient time for review. Please reach out to your academic advisor for more information or contact the DRC at drc@northeastern.edu

  Title IX and Sexual Misconduct: Title IX of the Education Amendments of 1972 protects indi- viduals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

  Northeastern University's Title IX Policy addresses Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty, and staff.

  If you or someone you know has been a survivor of a Prohibited Offense, confidential support and guidance can be found through University Health and Counseling Services staff (http://www.northeastern.edu/uhcs/) and the Center for Spiritual Dialogue and Service clergy mem- bers (http://www.northeastern.edu/spirituallife/). By law, those employees are not required to re- port allegations of sex or gender-based discrimination to the University.

  Alleged violations can be reported non-confidentially to the Title IX Coordinator within The Office for Gender Equity and Compliance at: titleix@northeastern.edu and/or through NUPD (Emergency 617.373.3333; Non-Emergency 617.373.2121). Reporting Prohibited Offenses to NUPD does NOT commit the victim/affected party to future legal action.

  Faculty members are considered "responsible employees" at Northeastern University, meaning they are required to report all allegations of sex or gender-based discrimination to the Title IX Coordi- nator.

  In case of an emergency, please call campus police.

Please visit http://www.northeastern.edu/titleix for a complete list of reporting options and re- sources both on- and off-campus.